

Lecture 22

Nesting and the Hubbard model

Objectives. Every model since [Lecture 15](#) has had one set of rapidities. Give the particles a spin and that stops being enough: the two-body S -matrix becomes a matrix on the labels, and carrying a particle round the ring acts on a vector instead of multiplying by a number. The quantisation condition is then an eigenvalue problem, and this lecture's one idea is that the eigenvalue problem is itself an integrable model — an inhomogeneous Heisenberg chain whose sites are the gas's own particles — solved by the algebraic Bethe ansatz of [Lecture 18](#). That brings a second set of rapidities and a second layer of Bethe equations. Then the Hubbard model, whose equations have the same shape with k replaced by $\sin k$.

22.1 Fermions with a spin, and a matrix S -matrix

The model is [Lecture 16](#)'s delta gas with one addition: each particle carries a spin- $\frac{1}{2}$ label, and the particles are fermions. On a ring of circumference L ,

$$H = - \sum_{j=1}^N \frac{\partial^2}{\partial x_j^2} + 2c \sum_{i < j} \delta(x_i - x_j), \quad c > 0, \quad (22.1)$$

acting on wavefunctions $\psi(x_1\sigma_1, \dots, x_N\sigma_N)$, $\sigma_j \in \{\uparrow, \downarrow\}$, that are *totally antisymmetric* under exchanging a position and its label together. This is the *Yang–Gaudin* model (Yang 1967, Gaudin 1967). Write M for the number of down spins, so $S^z = \frac{N}{2} - M$, and take $M \leq N/2$.

Remark 22.1 (Units, and two names used twice). Units are [Lecture 16](#)'s, $\hbar = 2m = 1$, kinetic operator $-\sum \partial_j^2$ — *not* the $\hbar = m = 1$ of [Lecture 21](#) ([Remark 21.1](#)). Two symbols are reused and are told apart by sight: the Hubbard hopping t of § 22.4 is a number while the transfer matrix $t(\lambda)$ always carries its argument, and Λ_α there is a spin rapidity while [Lecture 18](#)'s eigenvalue $\Lambda(\lambda)$ always carries its.

The two-body problem is two problems

[Lecture 16](#) did the two-body problem for labelled particles. On $x_1 < x_2$ write

$$\psi(x_1, x_2) = A e^{i(k_1 x_1 + k_2 x_2)} + A' e^{i(k_2 x_1 + k_1 x_2)}, \quad A, A' \in \mathbb{C}^2 \otimes \mathbb{C}^2, \quad (22.2)$$

with slot j of the tensor holding the label of the particle at position j , and extend by antisymmetry. [Proposition 16.6](#) gives $A' = Y_F(k_1 - k_2)A$ with

$$Y_F(k) = -\frac{kP + i c}{k + i c} = S(k) \Pi_- - \Pi_+, \quad S(k) = \frac{k - i c}{k + i c}, \quad (22.3)$$

P the permutation of the two label factors ([Definition 8.3](#)) and $\Pi_\pm = \frac{1}{2}(\mathbb{1} \pm P)$ the projectors on labels symmetric and antisymmetric · *checked*.

(22.3) is the whole reason this lecture exists. On the *singlet* — labels antisymmetric, positions symmetric — the delta acts and the amplitude picks up [Lecture 16](#)’s scalar $S(k)$. On the *triplet* the wavefunction vanishes where the particles meet, the delta never sees them, and the amplitude picks up -1 ([Figure 22.1](#)). One two-body problem, two answers; a scalar Bethe equation cannot hold both.

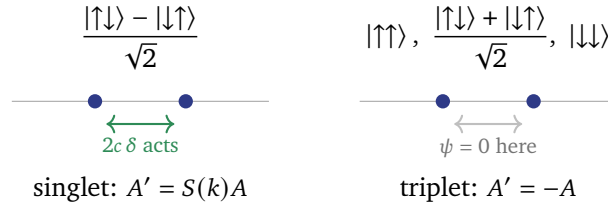


Figure 22.1: The two channels of (22.3) are *spin states of the colliding pair*, not configurations. Antisymmetric labels force symmetric positions, so the particles touch and the delta bites (left); symmetric labels force a wavefunction that vanishes where they meet, so the delta never sees them (right). A pair of opposite arrows is *not* the left-hand picture: $|\uparrow\downarrow\rangle = (|S\rangle + |T_0\rangle)/\sqrt{2}$ feels both amplitudes at once.

Following the momenta

[Lecture 16](#) also fixed the object to work with. Y is written in the order of *positions*; the natural bookkeeping for a scattering problem follows the *momenta*, and there the two-body matrix is $X = PY$ ([Equation \(16.14\)](#)), for fermions

$$X(u) = -\frac{u + iP}{u + i} = \frac{R(-u)}{u + i} = -\frac{L(u)}{u + i}, \quad u = \frac{k_1 - k_2}{c}, \quad (22.4)$$

with $R(u) = u - iP$ the course’s rational R-matrix and $L(u) = u + iP$ the Heisenberg Lax operator of [Lecture 17](#). We drop the subscript F: in this lecture X always means the fermionic one. Three facts, all checked symbolically, carry the lecture:

$$X(0) = -P, \quad X(u)X(-u) = \mathbb{1}, \quad X_{12}(u - v)X_{13}(u)X_{23}(v) = X_{23}(v)X_{13}(u)X_{12}(u - v). \quad (22.5)$$

The last is the Yang–Baxter equation (§ 16.6); the first is the only place a sign will come from in § 22.2. Note that (22.4) uses $R(-u)$, not R : in the course’s ordering of RTT ([Equation \(8.17\)](#)) the Lax operator pairs with $R(\lambda) = \lambda - iP$ and equals $\lambda + iP$, and a source with the other ordering carries its own R and its own L ([Remark 17.2](#)).

22.2 Periodicity is an eigenvalue problem

Inside the region $x_1 < x_2 < \dots < x_N$ the Bethe ansatz reads

$$\psi|_{x_1 < \dots < x_N} = \sum_{R \in \mathcal{S}_N} A_R e^{i \sum_j k_{R(j)} x_j}, \quad A_R \in (\mathbb{C}^2)^{\otimes N}, \quad (22.6)$$

slot j of A_R holding the label of the particle at position j ; total antisymmetry then determines ψ everywhere. The contact conditions at $x_j = x_{j+1}$ are [Lecture 16](#)’s walls, $A_{Rs_j} = Y_{j,j+1}(k_{R(j)} -$

$k_{R(j+1)}A_R$ (Equation (16.16)), and in the momentum-ordered amplitudes $B_R = \Pi_R A_R$ of Lemma 16.7 they become

$$B_{Rs_j} = X_{ab}(u_{ab})B_R, \quad a = R(j), \quad b = R(j+1), \quad u_{ab} = \frac{k_a - k_b}{c}. \quad (22.7)$$

Each crossing of momentum lines a and b multiplies by X_{ab} (Figure 22.2). Nothing is left to choose: the interaction has been used up. What remains is the ring.

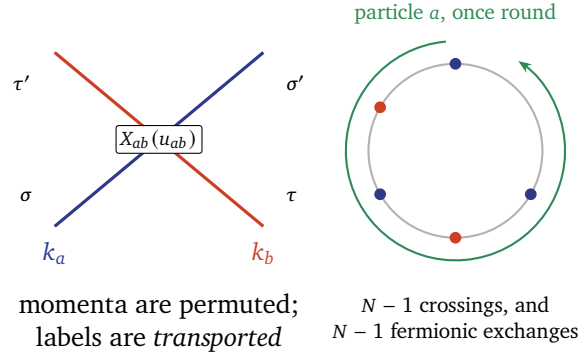


Figure 22.2: Left: in the momentum-ordered amplitudes a crossing is the matrix X_{ab} on the two labels — the momenta are permuted, as classically (Lecture 9), but the labels are carried across by an operator. Right: particle a once round the ring crosses every other particle once and moves past $N - 1$ others in the antisymmetric wavefunction. Both counts appear in Theorem 22.2.

Theorem 22.2 (Periodicity). *Let ψ be the totally antisymmetric extension of (22.6) with amplitudes satisfying (22.7). Then ψ is periodic in every variable if and only if, for every $R \in S_N$ and $a = R(1)$,*

$$(-1)^{N-1} e^{ik_a L} X_{a,R(N)}(u_{a,R(N)}) \cdots X_{a,R(2)}(u_{a,R(2)}) B_R = B_R. \quad (22.8)$$

Taking R to be the cyclic word $(a, a+1, \dots, N, 1, \dots, a-1)$, these are N conditions on a single vector $B = B_{\text{id}}$,

$$(-1)^{N-1} e^{ik_a L} \mathcal{X}_a B = B, \quad \mathcal{X}_a = X_{a,a-1} \cdots X_{a,1} X_{a,N} \cdots X_{a,a+1}, \quad (22.9)$$

every factor at its own argument $u_{ab} = (k_a - k_b)/c$.

Proof. Because ψ is antisymmetric, periodicity in one argument at every configuration gives it in all of them; take the first. Let $0 \leq x_1 < \dots < x_N < L$ and shift x_1 by L . The shifted point has its arguments in the order $x_2, \dots, x_N, x_1 + L$, so moving the first argument past the other $N - 1$ costs $(-1)^{N-1}$:

$$\psi(x_1 + L, \sigma_1; \dots) = (-1)^{N-1} \psi(x_2 \sigma_2, \dots, x_N \sigma_N, x_1 + L, \sigma_1),$$

and both sides of $\psi(x_1 + L, \dots) = \psi(x_1, \dots)$ are now values of (22.6). Matching the coefficient of each plane wave: the term R on the right contributes $e^{ik_{R(N)}L}$ and the momentum word $(R(N), R(1), \dots, R(N-1))$, while the slots of A_R are read in the cyclic order $(\sigma_2, \dots, \sigma_N, \sigma_1)$. Writing c for the cycle $1 \mapsto N, j \mapsto j-1$, and $C = \Pi_{c^{-1}}$ for the operator that permutes slots that way, the condition is

$$A_R = (-1)^{N-1} e^{ik_a L} C A_{Rc^{-1}}, \quad a = R(1). \quad (22.10)$$

Now pass to $B_R = \Pi_R A_R$. Since Π is a homomorphism, $\Pi_{Rc^{-1}}^{-1} = \Pi_c \Pi_R^{-1} = C^{-1} \Pi_R^{-1}$, so $CA_{Rc^{-1}} = \Pi_R^{-1} B_{Rc^{-1}}$ and the relabelling operator cancels: (22.10) is $B_R = (-1)^{N-1} e^{ik_a L} B_{Rc^{-1}}$. The permutation Rc^{-1} is R with its momentum word shifted one place left, reached from R by $s_1 s_2 \cdots s_{N-1}$, and at each step (22.7) applies with first leg a throughout: $B_{Rc^{-1}} = X_{a,R(N)} \cdots X_{a,R(2)} B_R$. That is (22.8).

The conditions for different R are not independent: those with the same a are conjugate to one another by the Yang–Baxter equation, and those with different a are carried onto the single vector B_{id} by a chain of the $\mathcal{X}$'s. Both reductions are below. ■

Proof of the reduction, and the two places it can go wrong (★). Same a . If $R' = Rs_j$ with $j \geq 2$, so that R and R' share $a = R(1)$ and differ by exchanging two letters b, b' of the word, then $B_{R'} = X_{bb'} B_R$, and the Yang–Baxter equation (22.5) in the form

$$X_{ab} X_{ab'} = X_{bb'} X_{ab} X_{ab}^{-1} X_{bb'}^{-1} \quad (22.11)$$

conjugates the whole product: the operator in (22.8) for R' is $X_{bb'}$ times the one for R times $X_{bb'}^{-1}$, because the remaining factors $X_{a,l}$ with $l \neq b, b'$ have legs disjoint from b, b' . So the two conditions are the same condition. **The order in (22.11) matters:** with the two $X_{bb'}$ exchanged the identity is false, and at $N = 3$ with $(\xi_a, \xi_b, \xi_{b'}) = (\frac{1}{3}, -\frac{5}{7}, \frac{2}{5})$ the two sides differ by an exact rational 0.43 in the largest entry · *checked*.

Different a . Every R with $R(1) = a$ is therefore equivalent to the cyclic word $C_a = (a, a + 1, \dots, N, 1, \dots, a - 1)$, with $C_1 = \text{id}$; but that is a condition on B_{C_a} , not on B_{id} . They are the same condition, by an induction on a . The word C_{a+1} is C_a with its first letter moved to the back, reached by $s_1 \cdots s_{N-1}$, and the crossings accumulated on the way are exactly $X_{a,C_a(N)} \cdots X_{a,C_a(2)} = \mathcal{X}_a$, so

$$B_{C_{a+1}} = \mathcal{X}_a B_{C_a}. \quad (22.12)$$

If the condition for a holds, $\mathcal{X}_a$ acts on B_{C_a} as the non-zero scalar $q_a = (-1)^{N-1} e^{-ik_a L}$, so $B_{C_{a+1}}$ is a non-zero multiple of B_{C_a} . Starting from $C_1 = \text{id}$, every B_{C_a} is a multiple of B and the N conditions are (22.9); conversely (22.12) runs the same induction backwards. All $N!$ conditions, and the chain, are verified at $N = 3$ and $N = 4$ · *checked*. ■

The fermionic sign in (22.9) is not cosmetic, and it is worth being precise about what dropping it does. At $N = 2$, where $\mathcal{X}_1 = X(u)$ and X acts on the singlet as $-S(k)$, the correct condition is $e^{ikL} = (u + i)/(u - i)$ and the one without the sign is $e^{ikL} = -(u + i)/(u - i)$, i.e. $\tan(\kappa L/2) = -2\kappa/c$ in place of $\tan(\kappa L/2) = c/2\kappa$ at zero total momentum. The root moves by a factor 2.007 at $L = 10, c = 3$ and 2.223 at $L = 7, c = 0.8$, the energy by 4.03 and 4.94 · *checked*. (That is a different mutation from using [Lecture 16's bosonic \$X\$](#) while keeping the fermionic sign, which gives $e^{ikL} = -1$; both are wrong and they are wrong differently.)

The product of crossings is a transfer matrix

(22.9) is N eigenvalue problems for N different operators, which looks hopeless. It is not: the N operators are values of one commuting family, and it is a family we have already solved.

Definition 22.3 (The auxiliary chain). Put $\xi_j = k_j/c$, adjoin an auxiliary $\mathbb{C}^2$ labelled a , and set

$$T_a(\lambda) = X_{aN}(\lambda - \xi_N) \cdots X_{a1}(\lambda - \xi_1), \quad t(\lambda) = \text{tr}_a T_a(\lambda). \quad (22.13)$$

Its quantum spaces are the *labels of the particles* and its N “sites” are the particles themselves, at inhomogeneities ξ_j equal to their own momenta in units of the coupling. There is

no lattice and no spin Hamiltonian: the chain is a bookkeeping device that happens to be an integrable model.

Proposition 22.4 (The crossings are the transfer matrix). For every j , $t(\xi_j) = -\mathcal{X}_j$.

Proof. At $\lambda = \xi_j$ the j th factor of (22.13) is $X_{aj}(0) = -P_{aj}$:

$$T_a(\xi_j) = -X_{aN} \cdots X_{a,j+1} P_{aj} X_{a,j-1} \cdots X_{a1}.$$

Push P_{aj} to the right using $P_{aj}X_{al} = X_{jl}P_{aj}$ for $l \neq j$; the factors to the right of it become operators on the quantum spaces alone, $\mathcal{M} = X_{j,j-1} \cdots X_{j1}$, which involves only the spaces $1, \dots, j$ and so commutes with $X_{aN} \cdots X_{a,j+1}$. Hence $T_a(\xi_j) = -\mathcal{M} X_{aN} \cdots X_{a,j+1} P_{aj}$, and $\text{tr}_a(F_a P_{aj}) = F_j$, valid for any F not acting on space j , gives $t(\xi_j) = -\mathcal{M} X_{jN} \cdots X_{j,j+1} = -\mathcal{X}_j$. ■

Proposition 22.5 (The chain is Heisenberg). $T_a(\lambda) = (-1)^N \left[\prod_n (\lambda - \xi_n + i) \right]^{-1} L_{aN}(\lambda - \xi_N) \cdots L_{a1}(\lambda - \xi_1)$: the auxiliary chain is the inhomogeneous Heisenberg chain of [Lecture 18](#)'s reading, up to one scalar. Consequently $[t(\lambda), t(\mu)] = 0$ for all λ, μ , and $[t(\lambda), S^z] = 0$.

Proof. The displayed identity is (22.4) factor by factor. The *RTT* relation ([Theorem 17.3](#)) needs only that every site operator obeys *RLL* with one common R , and the shifts cancel in the argument of R , $(\lambda - \xi_n) - (\mu - \xi_n) = \lambda - \mu$, so commutativity follows as in [Lecture 17](#); scalars commute with everything · *checked*. ■

So all N conditions can be imposed at once: ask for a common eigenvector of the family $t(\lambda)$ and read off the N numbers $t(\xi_a)$. *This is the whole of nesting.* The first Bethe ansatz reduced the gas to a finite linear-algebra problem in the labels; the second solves that problem.

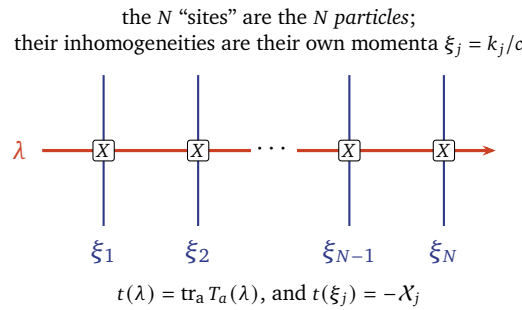


Figure 22.3: The auxiliary chain of [Definition 22.3](#), drawn as [Lecture 17](#) drew the Heisenberg chain — with the one difference that is the commonest place to stall: its sites are not lattice sites but the particles of a gas on a continuous ring, each at its own rapidity. Setting λ equal to one of them collapses the horizontal line onto a vertical one ([Proposition 22.4](#)), which is why the commuting family contains the N periodicity operators.

22.3 The nested Bethe equations

[Lecture 18](#) solved this chain. Work with the *unnormalised* monodromy of [Proposition 22.5](#),

$$\tilde{T}(\lambda) = L_{aN}(\lambda - \xi_N) \cdots L_{a1}(\lambda - \xi_1) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (22.14)$$

whose entries are the A, B, C, D below — *not* those of (22.13), which differ from them by the scalar and would give $A|0\rangle = (-1)^N|0\rangle$. Its reference state is all labels up, $|0\rangle$, on which $\tilde{T}$ is triangular with $A(\lambda)|0\rangle = \prod_n(\lambda - \xi_n + i)|0\rangle$ and $D(\lambda)|0\rangle = \prod_n(\lambda - \xi_n)|0\rangle$ (Proposition 18.1 with $\lambda \mapsto \lambda - \xi_n$); B creates; and for pairwise distinct $v_1, \dots, v_M$, none equal to 0 or $-i$, the vector $\Psi = B(v_1) \cdots B(v_M)|0\rangle$ is an eigenvector of the whole family, with

$$\Lambda(\lambda) = \prod_n (\lambda - \xi_n + i) \prod_\alpha \frac{\lambda - v_\alpha - i}{\lambda - v_\alpha} + \prod_n (\lambda - \xi_n) \prod_\alpha \frac{\lambda - v_\alpha + i}{\lambda - v_\alpha}, \quad (22.15)$$

provided the residues at $\lambda = v_\alpha$ cancel (Corollary 18.6).

Remark 22.6 (Which shift, and why it is not free). Write the spin rapidities as $\lambda_\alpha = v_\alpha + \frac{i}{2}$. Lecture 18's homogeneous chain names the same roots $v_\alpha = -\lambda_\alpha - \frac{i}{2}$ (Equation (18.5)), so *this lecture's* λ_α is *minus* Lecture 15's. There the sign was a labelling: the homogeneous system is invariant under $\lambda \rightarrow -\lambda$ and the two choices name a mirror doublet, which only the *vector* tells apart. Here the ambiguity does not arise — with inhomogeneities the system is not invariant and the mirrored roots do not solve it — but the check script builds $B(v_1) \cdots B(v_M)|0\rangle$ anyway and confirms it is the eigenvector used — *checked*.

Theorem 22.7 (The nested Bethe equations, Yang and Gaudin 1967). Let $k_1, \dots, k_N$ be pairwise distinct with no two differing by $\pm ic$ (so that no X in (22.9) is singular), let $\lambda_1, \dots, \lambda_M$ be pairwise distinct and $\neq \pm \frac{i}{2}$ (Corollary 18.6's regular roots, $v_\alpha \neq 0, -i$), put $\xi_j = k_j/c$, and let

$$e^{ik_j L} = \prod_{\alpha=1}^M \frac{\xi_j - \lambda_\alpha + \frac{i}{2}}{\xi_j - \lambda_\alpha - \frac{i}{2}}, \quad j = 1, \dots, N, \quad (22.16)$$

$$\prod_{j=1}^N \frac{\lambda_\alpha - \xi_j + \frac{i}{2}}{\lambda_\alpha - \xi_j - \frac{i}{2}} = \prod_{\beta \neq \alpha}^M \frac{\lambda_\alpha - \lambda_\beta + i}{\lambda_\alpha - \lambda_\beta - i}, \quad \alpha = 1, \dots, M. \quad (22.17)$$

Then $\Psi = B(\lambda_1 - \frac{i}{2}) \cdots B(\lambda_M - \frac{i}{2})|0\rangle$, if it is not zero, satisfies every periodicity condition (22.9), and the resulting eigenfunction of (22.1) has

$$E = \sum_{j=1}^N k_j^2, \quad K = \sum_{j=1}^N k_j, \quad S^z = \frac{N}{2} - M. \quad (22.18)$$

Proof. (22.17) is Corollary 18.6's system (18.16) for the inhomogeneous chain, written in $\lambda_\alpha = v_\alpha + \frac{i}{2}$: the residue of (22.15) at v_α vanishes exactly when $\prod_n (v_\alpha - \xi_n + i) / \prod_n (v_\alpha - \xi_n) = \prod_{\beta \neq \alpha} (v_\alpha - v_\beta + i) / (v_\alpha - v_\beta - i)$. So Ψ is a common eigenvector and Proposition 22.4 turns (22.9) into N scalar equations, $(-1)^{N-1} e^{ik_\alpha L} \cdot (-\Lambda^X(\xi_\alpha)) = 1$, with Λ^X the eigenvalue of (22.13). Evaluate it. By Proposition 22.5, $\Lambda^X(\lambda) = (-1)^N \Lambda(\lambda) / \prod_n (\lambda - \xi_n + i)$; at $\lambda = \xi_\alpha$ the second term of (22.15) carries the factor $\prod_n (\xi_\alpha - \xi_n) = 0$ and dies, and the first term's prefactor cancels the denominator *exactly*, leaving

$$\Lambda^X(\xi_\alpha) = (-1)^N \prod_\alpha \frac{\xi_\alpha - \lambda_\alpha - \frac{i}{2}}{\xi_\alpha - \lambda_\alpha + \frac{i}{2}}. \quad (22.19)$$

Then $(-1)^{N-1} \cdot (-1) \cdot (-1)^N = 1$ for every N , and what is left is (22.16). Inside the sectors every term of (22.6) is a plane wave of energy $\sum k_j^2$ and momentum $\sum k_j$, as in Theorem 16.3. ■

Two features deserve to be said aloud.

The two sets do different jobs. The k_j are momenta and carry the energy (22.18). The λ_α are dimensionless, as many as there are down spins, and carry none of it — they enter (22.16) only as phase shifts. That is spin–charge separation in its rawest form, and the displacement is not small (Figure 22.4).

The counting. Taking logarithms of (22.16)–(22.17) with the identities $(x \pm \frac{i}{2})$, $(x \pm i)$ written as arctangents gives

$$k_j L + 2 \sum_{\alpha} \arctan(2(\xi_j - \lambda_{\alpha})) = 2\pi I_j, \quad \sum_j \arctan(2(\lambda_{\alpha} - \xi_j)) - \sum_{\beta \neq \alpha} \arctan(\lambda_{\alpha} - \lambda_{\beta}) = \pi J_{\alpha}, \quad (22.20)$$

with $I_j \in \mathbb{Z} + \frac{M}{2}$ and $J_{\alpha} \in \mathbb{Z} + \frac{N-M+1}{2}$, and $K = (2\pi/L)(\sum_j I_j + \sum_{\alpha} J_{\alpha})$. **The parities are part of the equations:** a half-integer I where an integer belongs solves (22.16) with an extra minus sign, and no residual of (22.20) can see it. Every solution must be checked in the product form · checked.

Example 22.8 ($N = 2$: the two channels, again). Take $N = 2$, $M = 1$. (22.17) reads $(\lambda - \xi_1 + \frac{i}{2})(\lambda - \xi_2 + \frac{i}{2}) = (\lambda - \xi_1 - \frac{i}{2})(\lambda - \xi_2 - \frac{i}{2})$, whose only root is $\lambda = \frac{1}{2}(\xi_1 + \xi_2)$; then $\xi_1 - \lambda = \frac{1}{2}(\xi_1 - \xi_2)$ and (22.16) becomes

$$e^{ik_1 L} = \frac{k_1 - k_2 + ic}{k_1 - k_2 - ic},$$

exactly Lecture 16’s Lieb–Liniger equation (16.9) for two particles. The other $M = 1$ state is the descendant $\lambda \rightarrow \infty$, where the right-hand side of (22.16) tends to 1: free fermions. The two channels of Figure 22.1, recovered from the nesting · checked.

At zero total momentum the same root comes out of a problem with no Bethe ansatz in it at all — a delta well on a circle — which the tutorial does · checked.

Whether these solutions exhaust the spectrum is a separate question, and the answer is § 15.6’s: they do not, without the complex ones. The reading at the end says what is known · cited: Takahashi ch. 12; Essler et al. ch. 4.

22.4 The Hubbard model

What is proved in this section, and what is not. The Hubbard model’s Bethe equations are stated; their derivation, which is a nested algebraic Bethe ansatz over an R -matrix that this course has not built, is cited. What is done here: the two-body problem is solved on the lattice, in the tutorial, and it fixes the correspondence with § 22.3 including the sign; the equations are then checked against exact diagonalisation of a six-site ring, where 20 of the 21 states of the singlet sector are regular Bethe states and the twenty-first is the η -pair of Remark 22.10. Eighteen of the twenty — those with $K \neq \pi$, complex roots included — are checked to 10^{-14} by reconstructing the roots from (K, E) ; that reconstruction degenerates at $K = \pi$, so the two states there are checked directly against the equations instead · checked. The $SO(4)$ commutators are computed exactly, on one even ring.

On a ring of L sites — L even, which Remark 22.10 needs and nothing before it does — with $c_{j\sigma}^{\dagger}$ creating an electron of spin σ at site j and $n_{j\sigma} = c_{j\sigma}^{\dagger} c_{j\sigma}$,

$$H = -t \sum_{j=1}^L \sum_{\sigma} (c_{j\sigma}^{\dagger} c_{j+1,\sigma} + c_{j+1,\sigma}^{\dagger} c_{j\sigma}) + U \sum_{j=1}^L n_{j\uparrow} n_{j\downarrow}. \quad (22.21)$$

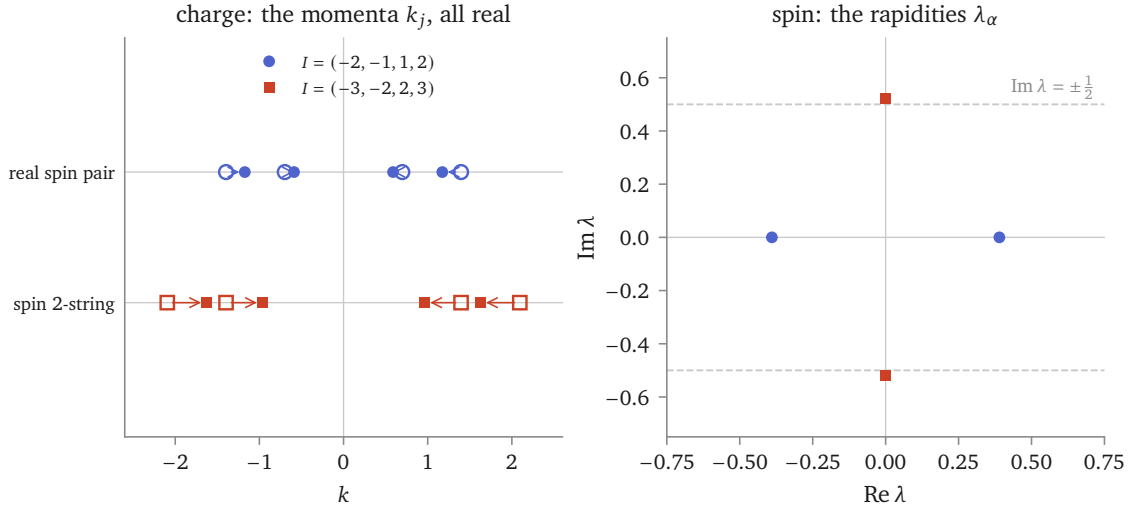


Figure 22.4: Two spin-singlet Bethe states at $N = 4$, $M = 2$, $L = 9$, $c = 3$, total momentum zero. Left: both have four *real* momenta, so this panel is one axis and the two rows are only a device for keeping the states apart. Each is drawn against its own $2\pi I_j/L$ (open marks, arrow the shift), and the shift is inward for both. **The two states do not share their charge quantum numbers:** (22.20) gives $(-2, -1, 1, 2)$ for one and $(-3, -2, 2, 3)$ for the other, which no residual of (22.16) can see. Right: one state has a real spin pair, the other a 2-string on the imaginary axis — a bound state of two spin waves — with energy 7.17 against 3.44. From `scripts/figures/nested_roots.py`.

Only U/t matters, and we set $t = 1$ from here. It is the Yang–Gaudin gas put on a lattice: two species, and an interaction that acts only when they meet. Two things change at once — the dispersion becomes $-2 \cos k$ instead of k^2 , and the delta becomes an on-site energy — and that the two conspire into the single substitution (22.24) is a result, not an observation. The dispersion is what makes it hard: $-2 \cos k$ is not a quadratic form, so there is no Galilean invariance and no centre-of-mass separation.

Theorem 22.9 (Lieb and Wu 1968). *The regular eigenstates of (22.21) with N electrons and M down spins — those with finite, pairwise distinct roots, and $2M \leq N \leq L$ — are given by charge momenta $k_1, \dots, k_N$ and spin rapidities $\Lambda_1, \dots, \Lambda_M$ solving*

$$e^{ik_j L} = \prod_{\alpha=1}^M \frac{\sin k_j - \Lambda_\alpha + iU/4}{\sin k_j - \Lambda_\alpha - iU/4}, \quad (22.22)$$

$$\prod_{j=1}^N \frac{\Lambda_\alpha - \sin k_j + iU/4}{\Lambda_\alpha - \sin k_j - iU/4} = \prod_{\beta \neq \alpha}^M \frac{\Lambda_\alpha - \Lambda_\beta + iU/2}{\Lambda_\alpha - \Lambda_\beta - iU/2}, \quad (22.23)$$

with $E = -2 \sum_j \cos k_j$ · cited: Lieb and Wu (1968); Essler et al. ch. 3. **They are not all the states**, and not all the rest are symmetry descendants either: at isolated couplings a pair of charge momenta collides and leaves a state that is neither. The reading gives one, on the six-site ring. Completeness is a counting statement with known exceptions and a subject of its own · cited: Essler et al. §3.2.6, ch. 4.

Compare (22.22)–(22.23) with (22.16)–(22.17), written in physical units with $\Lambda_\alpha = c\lambda_\alpha$:

they are the same equations under

$$k_j \mapsto \sin k_j, \quad c \mapsto \frac{U}{2t}. \quad (22.24)$$

The lattice replaces the momentum by its sine, and the coupling by $U/2t$ — with *that* sign. Getting it wrong is not a convention: with $c \mapsto -U/2t$ not one of the eighteen singlet states of the two-electron $L = 6$ ring satisfies (22.22), and with the right sign all eighteen do · *checked*. The tutorial at the end derives (22.24) from the lattice cusp condition.

Remark 22.10 ($SO(4)$). (22.21) has more symmetry than the spin $su(2)$. Let $\eta^\dagger = \sum_j (-1)^j c_{j\uparrow}^\dagger c_{j\downarrow}^\dagger$ create a pair on a site, with the staggering. Then

$$[H, \eta^\dagger] = U\eta^\dagger, \quad [H, S^+] = 0, \quad [\eta^\dagger, S^+] = 0,$$

all three for **even** L : the hopping terms cancel in pairs along the ring, and the closing bond cancels only if $(-1)^L = 1$. At $L = 3$, $U = 7$ the commutator misses $U\eta^\dagger$ by 2 · *checked*. So $H - \frac{U}{2}\hat{N}$ carries a second $su(2)$ commuting with the first. As Lie algebras $su(2) \oplus su(2) \cong so(4)$; the $\mathbb{Z}_2$ of the familiar statement $(SU(2) \times SU(2))/\mathbb{Z}_2 \cong SO(4)$ lives at the group level, and no commutator can see it.

For the printed H the η 's are spectrum-generating, not symmetries: $[H, \eta^\dagger] = U\eta^\dagger$ moves the energy rather than fixing it, and it is $H - \frac{U}{2}\hat{N}$ whose degeneracies $SO(4)$ organises · *cited: Yang (1989); Essler et al. ch. 2*. The staggering makes $\eta^\dagger$ anticommute with the translation, so it raises the momentum by π , and $\eta^\dagger |0\rangle$ is an exact eigenstate of (22.21) of energy U at $K = \pi$ · *checked* — and outside Theorem 22.9, because at $K = \pi$ every finite Bethe pair has $\cos k_1 + \cos k_2 = 0$, so $E = 0$, while exact diagonalisation at $L = 6$ returns 0, 0 and U .

There is an R -matrix behind (22.22)–(22.23) — Shastry's, which obeys the Yang–Baxter equation but is **not of difference form**, depending on the two spectral parameters separately · *cited: Shastry (1986); Essler et al. ch. 12*. Every construction in Lecture 17 and Lecture 18 used the difference property. So the Hubbard model is solvable by these methods and is not an instance of them, which is the honest place to leave it in a one-term course; the reading at the end says what Shastry's matrix is, and what it is *not* the reason for. Lecture 23 takes the nested equations (22.16)–(22.17) and (22.22)–(22.23) as given and asks what a gas described by two coupled families of rapidities does out of equilibrium.

22.5 Higher rank: where the Dynkin diagram comes in

Nothing in § 22.2 used that the label space is $\mathbb{C}^2$. For κ species the two-body matrix is still $X(u) = R(-u)/(u + i)$ with $R(u) = u - iP$ on $\mathbb{C}^\kappa \otimes \mathbb{C}^\kappa$, it still obeys the Yang–Baxter equation · *checked*, and Theorem 22.2 and Proposition 22.4 go through word for word. What changes is § 22.3. The auxiliary *monodromy* is now $\kappa \times \kappa$ in the auxiliary index — the transfer matrix $t = \text{tr}_a T_a$ still acts on the κ^N -dimensional label space — its reference state breaks sl_κ to $sl_{\kappa-1}$, and its first row carries $\kappa - 1$ creation operators $B_2(\lambda), \dots, B_\kappa(\lambda)$ in place of Lecture 18's single B . A Bethe vector $B_{i_1}(\lambda_1) \cdots B_{i_M}(\lambda_M) |0\rangle$ therefore carries a vector of indices, and choosing that vector is itself a Bethe-ansatz problem of rank $\kappa - 1$, whose auxiliary problem needs another, and so on — $\kappa - 1$ levels, the last of them scalar. That is why the procedure is called nesting, and the extra creation operators are what Lecture 18 pointed at.

The bookkeeping is the Lie algebra's. $R(u) = u - iP$ commutes with $A \otimes 1 + 1 \otimes A$ for every $A \in sl_\kappa$ · *checked*, so the hierarchy is sl_κ -invariant, and the $\kappa - 1$ levels are the nodes of the

Dynkin diagram $A_{\kappa-1}$: level a carries M_a rapidities $\lambda^{(a)}$, and the equations couple level a to level b through the Cartan matrix,

$$\prod_{j=1}^N \frac{\lambda_\alpha^{(a)} - \xi_j + \frac{i}{2}\delta_{a1}}{\lambda_\alpha^{(a)} - \xi_j - \frac{i}{2}\delta_{a1}} = \prod_{b=1}^{\kappa-1} \prod_{\beta \neq (a,\alpha)} \frac{\lambda_\alpha^{(a)} - \lambda_\beta^{(b)} + iC_{ab}/2}{\lambda_\alpha^{(a)} - \lambda_\beta^{(b)} - iC_{ab}/2}, \quad C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \text{ for } A_2, \quad (22.25)$$

the source appearing only at the node the particles attach to — the left-hand side is 1 at every other node, $\delta_{a1} = 0$ — and the term $(b, \beta) = (a, \alpha)$ **excluded**: with $C_{aa} = 2$ it would contribute $i/(-i) = -1$ and turn the $\kappa = 2$ case into $1 = -1$ · *checked*. Equation (22.17) is that $\kappa = 2$ case (Figure 22.5); the A_2 pair, whose second equation contains no ξ at all, is in the reading · *cited*: Sutherland (1968), (16)–(17); Babelon, de Vega and Viallet (1982).

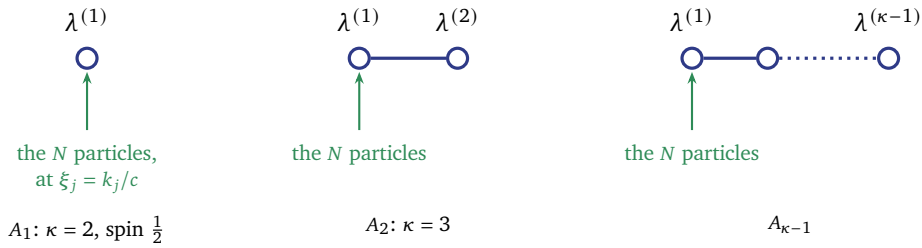


Figure 22.5: The levels of the nesting are the nodes of the Dynkin diagram and the gas hangs off the first. Each node carries its own rapidities $\lambda^{(a)}$; nodes joined by an edge have $C_{ab} = -1$ and appear in each other's equations, unjoined ones have $C_{ab} = 0$ and do not. For $\kappa = 2$ there is one node, (22.17). Only that first node sees the ξ_j .

The same diagram indexes the other series once the label space carries a different representation, and then the representation data — weights, matrix elements of the generators — is what the computation needs; the `rep-engine` package supplies it exactly, and the check script takes sl_3 's fundamental from it.

This is where Lecture 12's reading returns: there the root systems A_{N-1} , B_N , C_N , D_N indexed *classical* Calogero–Moser models, one interaction per positive root; here $A_{\kappa-1}$ indexes the levels of a *quantum* Bethe ansatz, one set of rapidities per node. The same combinatorial object, two unrelated uses, and no theorem in this course connecting them.

Dictionary

The root system of sl_N , indexing the classical Calogero–Moser models — one pair interaction per positive root (Lecture 12, reading) → the Dynkin diagram of $A_{\kappa-1}$, indexing the levels of the nested Bethe ansatz for κ -component particles — one set of rapidities per node, coupled by the Cartan matrix (22.25) (type A, *cited*)

Elastic non-diffractive scattering: no *new* asymptotic momenta appear, the set is only permuted, and what a collision leaves behind is a displacement (Lecture 9, export E6; the separation itself is Theorem 9.9, *cited*) → the momenta are still only permuted (Equation (22.18)) and the collision still leaves a phase, but the labels are *transported* by X (Figure 22.2), so the ring condition is an eigenvalue problem and not a number (type A; the Part II side proved, the Part I side *cited*)

22.6 Shastry's R-matrix, and two transcription traps

The R-matrix. Shastry (1986) built a two-dimensional vertex model whose transfer matrix commutes with (22.21), by taking two six-vertex models at their free-fermion point and coupling them with a *local* diagonal factor $\exp(\hbar C^\dagger C^\downarrow/2)$ whose parameter is tied to the spectral one by $\sinh 2h = u \sin 2z$. Two things are worth separating. Jordan–Wigner is the transformation between the fermions and the spin variables, not the coupling. And the loss of difference form is *not* what stops t at a special point from being the shift operator: that is *regularity*, $R(z, z) \propto P$, which Shastry's matrix has. What difference form buys is one R generating a whole commuting family from a single argument, which is what [Lecture 17](#)'s construction used · *cited: Shastry (1986); Essler et al. ch. 12.*

The other nodes. For A_2 the second node of (22.25) reads

$$1 = \prod_{\beta \neq \alpha} \frac{\lambda_\alpha^{(2)} - \lambda_\beta^{(2)} + i}{\lambda_\alpha^{(2)} - \lambda_\beta^{(2)} - i} \prod_\gamma \frac{\lambda_\alpha^{(2)} - \lambda_\gamma^{(1)} - \frac{i}{2}}{\lambda_\alpha^{(2)} - \lambda_\gamma^{(1)} + \frac{i}{2}},$$

with no ξ in it at all: the second level sees the particles only through the first. Sutherland's original writes both levels with an external minus and the self-root included, which is the same equation once the self-factor -1 is taken across · *cited: Sutherland (1968), (16)–(17).*

The two statistics. The *bosonic* $X_B = R(u)/(u+i)$ of [Lecture 16](#) also has $X_B(0) = -P$, because $R(0) = -iP$: the two statistics are told apart by $u \rightarrow -u$, not by the value at the origin, and [Proposition 22.4](#) holds for both. What they do not share is the periodicity condition · *checked.*

An exceptional state. [Theorem 22.9](#) is restricted to regular solutions, and its escape clause is not decorative. At $L = 6$, $N = 2$, $U = 2\sqrt{3}/3$ the symmetric wavefunction $e^{ik_0(x+y)}(3 - |x - y|)$ with $k_0 = 7\pi/6$ is an exact eigenstate of energy $2\sqrt{3}$ at $K = \pi/3$. Any two-root representation of it must satisfy $E = -4 \cos(K/2) \cos \frac{1}{2}(k_1 - k_2)$, which here forces $k_1 = k_2$: at coincident roots the cusp condition gives $A' = -A$ and the ordinary ansatz vanishes, so this state is a regularised *limit* of a colliding pair, reached as $q \rightarrow 0$ along $k = k_0 \pm iq$. Nor is it a descendant: η annihilates it and the spin generators preserve the total spin · *checked.*

The Hamiltonian. The reference book writes (22.21) in its particle–hole symmetric form, $U \sum (n_\uparrow - \frac{1}{2})(n_\downarrow - \frac{1}{2})$, so its energy carries an extra $\frac{U}{4}(L - 2N)$; its coupling is $u = U/4t$, ours is written out; and its spin equation is the reciprocal of (22.23). All three differences are invisible in a formula copied out of context · *checked.*

The gap. § 22.7's Δ_μ is the chemical-potential discontinuity. The same book's Δ is the single-holon threshold, exactly half of it, and is printed in Bessel form with $u = U/4t$. Which gap is meant is not a detail: the two differ by a factor of two at every coupling · *checked.*

Reading

22.7 Half filling, the Mott gap, and spin–charge separation

At half filling, $N = L$, the Hubbard model is an insulator for every $U > 0$. Lieb and Wu’s own paper established it from (22.22)–(22.23) in the thermodynamic limit: the *charge gap*, meaning the discontinuity $\mu_+ - \mu_-$ of the chemical potential, is

$$\Delta_\mu = \frac{16t^2}{U} \int_1^\infty \frac{\sqrt{y^2 - 1}}{\sinh(2\pi ty/U)} dy,$$

positive for every $U > 0$, tending to $U - 4t$ from above and to $(8/\pi)\sqrt{U}t e^{-2\pi t/U}$ as $U \rightarrow 0$ · *cited: Lieb and Wu (1968), (21)–(23)*. Say which gap. The reference book’s Δ (its (7.18)) is the *single-holon* threshold, exactly half of this, and is written in Bessel form with $u = U/4t$; the two agree to every digit once converted · *checked*. There is no metal–insulator transition at finite U in one dimension; the transition is at $U = 0^+$. The spin sector meanwhile stays gapless, and at large U the half-filled model reduces to the Heisenberg antiferromagnet of Lecture 15 with $J = 4t^2/U$: second-order hopping of an electron onto an occupied site and back costs U and gains t^2 , and it can only happen if the two spins are antiparallel. On a four-site ring, where the Heisenberg ground energy is $-3J = -12/U$, the two agree to 0.62% at $U = 40$ and 0.039% at $U = 160$ · *checked*.

That is spin–charge separation, and it is visible already in § 22.3: the energy (22.18) does not contain λ_α . In the thermodynamic limit the two sectors acquire different sound velocities — v_c from the k ’s and v_s from the λ ’s — and an added electron’s charge and spin propagate apart. The Luttinger-liquid description of the resulting low-energy theory, and the angle-resolved photoemission experiments on quasi-one-dimensional conductors that look for the two dispersing features, are the obvious continuation · *cited: Essler et al. ch. 8; Giamarchi, Quantum Physics in One Dimension*.

Reading

22.8 What is not claimed about the gas

That the solutions of (22.16)–(22.17) exhaust the spectrum is *not* proved here, and the situation is the one § 15.6 described for the Heisenberg chain: real roots do not suffice. Here the complex ones live in the *spin* sector — strings of λ ’s, one of which is drawn in Figure 22.4 — while for $c > 0$ the momenta k_j stay real. Proposition 16.4 proves that for the *spinless* gas; for the *spinful* one it is Yang’s and Gaudin’s statement, and it is not proved here. Completeness is a counting statement with known exceptions · *cited: Takahashi ch. 12; Essler et al. ch. 4*. The thermodynamics built on that string hypothesis — the analogue of Lecture 20 for two coupled families — is a genuine subject and is not attempted.

22.9 Bosons with a label, and the attractive gas

Two variants cost one line each and are worth knowing.

Two-component bosons. (22.3) becomes $Y_B = S\Pi_+ - \Pi_-$, so $X_B(u) = R(u)/(u + i)$ — the R -matrix itself rather than $R(-u)$, and the interacting channel is now the symmetric one. The auxiliary chain is again of Heisenberg type, the fermionic $(-1)^{N-1}$ of (22.9) is absent, and the nested equations come out with the shifts reflected. Substituting the bosonic X into the fermionic condition without touching the statistics sign gives $e^{ikL} = -1$ in the singlet: antiperiodic quantisation, and a spectrum wrong by half a level · *checked*.

Attraction. For $c < 0$ the k_j need no longer be real (Proposition 16.4 assumed $c > 0$), and the gas forms bound pairs of one up and one down spin — the one-dimensional analogue of a Cooper pair. In the *thermodynamic limit* the pair is the string $k = c\lambda \pm ic/2$ · *cited*: Gaudin, The Bethe Wavefunction, §11.4, where the string is presented as a large-system approximation. It is not exact on a finite ring: at $N = 2$, $K = 0$, $c = -g < 0$ the even bound wave is $\cosh q(x - L/2)$ with $2q \tanh(qL/2) = g$, so $q > g/2$; at $g = 1$, $L = 2$ the true q is 0.7717 against the string's 0.5 · *checked*. Nothing in § 22.2 used the sign of c ; everything in § 22.3 about *real* solutions did.

22.10 Two electrons on a lattice, from scratch

The Lieb–Wu equations were stated, not derived. Here is the $N = 2$ case in full, on a lattice of L sites, and it is enough to fix (22.24) including its sign.

Take one up electron and one down electron; the spin singlet has a symmetric spatial wavefunction $\psi(x_1, x_2)$. For $x_1 \leq x_2$ put

$$f(x_1, x_2) = A e^{i(k_1 x_1 + k_2 x_2)} + A' e^{i(k_2 x_1 + k_1 x_2)},$$

set $\psi = f$ there and $\psi(x_1, x_2) = f(x_2, x_1)$ elsewhere. Away from the diagonal the lattice Schrödinger equation is free and gives $E = -2(\cos k_1 + \cos k_2)$ for any A, A' .

The diagonal. At $x_1 = x_2 = x$ the equation reads

$$-[\psi(x-1, x) + \psi(x+1, x) + \psi(x, x-1) + \psi(x, x+1)] + U\psi(x, x) = E\psi(x, x),$$

and by symmetry $\psi(x+1, x) = f(x, x+1)$ and $\psi(x, x-1) = f(x-1, x)$, while the free equation for f continued to the diagonal is the same expression with every ψ replaced by f and no U . Subtracting,

$$[f(x+1, x) + f(x, x-1) - f(x-1, x) - f(x, x+1)] + Uf(x, x) = 0.$$

Every term carries $e^{i(k_1 + k_2)x}$; cancelling it leaves a linear relation whose coefficients are, with $s_j = \sin k_j$,

$$[2i(s_1 - s_2) + U]A + [-2i(s_1 - s_2) + U]A' = 0, \quad \text{so} \quad \frac{A'}{A} = \frac{s_1 - s_2 - iU/2}{s_1 - s_2 + iU/2}$$

· *checked*. That is Lecture 16's $S(k) = (k - ic)/(k + ic)$ with $k \mapsto s_1 - s_2$ and $c \mapsto U/2$ — (22.24), and its sign, derived rather than transcribed.

The ring. Periodicity relates the two amplitudes once more, and here it is worth doing rather than quoting: the antisymmetry sign of [Theorem 22.2](#) does not appear, because we are working with the *symmetric spatial* function of the singlet. Since $x_2 < x_1 + L$, $\psi(x_1+L, x_2) = f(x_2, x_1+L)$, so $f(x_2, x_1+L) = f(x_1, x_2)$; matching the coefficients of $e^{i(k_2x_1+k_1x_2)}$ and $e^{i(k_1x_1+k_2x_2)}$ gives

$$A e^{ik_2L} = A', \quad A' e^{ik_1L} = A, \quad \text{so} \quad e^{ik_1L} = \frac{A}{A'} = \frac{s_1 - s_2 + iU/2}{s_1 - s_2 - iU/2},$$

which is (22.22) at $M = 1$ once (22.23) has been used: at $M = 1$ its right-hand side is 1 and its left-hand side forces $\Lambda = \frac{1}{2}(s_1 + s_2)$, so $s_1 - \Lambda = \frac{1}{2}(s_1 - s_2)$ and the two displayed forms agree. Multiplying the two charge equations gives $e^{i(k_1+k_2)L} = 1$: the total momentum is quantised, as it must be.

The test. Diagonalise (22.21) on $L = 6$ with one up and one down electron — a 36×36 matrix — and resolve it by total momentum and by spatial symmetry. There are 21 symmetric states. Eighteen of them, all those with $K \neq \pi$, satisfy the displayed equation to 10^{-14} at $U = 2, 5$ and 12 ; some need complex k_1, k_2 with $k_1 + k_2 = K$, which is what a bound k - Λ pair looks like · *checked*. The other three sit at $K = \pi$, where the energy $-2(\cos k_1 + \cos k_2)$ vanishes for *every* finite pair with $k_1 + k_2 = \pi$ and so cannot fix $k_1 - k_2$: there the equation has to be read directly. It gives $e^{ik_1L} = -1$, because $k_2 = \pi - k_1$ makes $s_1 = s_2$ and the ratio $(iU/2)/(-iU/2)$, so k lies on the *antiperiodic* grid $(2m+1)\pi/6$ — the pairs $\{\pi/6, 5\pi/6\}$ with $\Lambda = \frac{1}{2}$ and $\{-\pi/6, 7\pi/6\}$ with $\Lambda = -\frac{1}{2}$, which are the two $E = 0$ states and are perfectly regular solutions · *checked*; only the (K, E) route to them fails. **It is not true that any real pair summing to π solves it:** $(0, \pi)$ gives $e^{ik_1L} = 1$ against -1 . The third state, at $E = U$, is the η -pair of [Remark 22.10](#) and is outside [Theorem 22.9](#) altogether.

The continuum check, for comparison. [Example 22.8](#)'s $N = 2$ gas can be solved without any Bethe ansatz. At zero total momentum, $k_1 = -k_2 = \kappa$, the relative coordinate obeys $-2\varphi'' + 2c\delta\varphi = E\varphi$ on a circle of circumference L with φ even, so $\varphi = \cos \kappa(x - L/2)$ and the jump condition $2\varphi'(0^+) = c\varphi(0)$ gives $\tan(\kappa L/2) = c/2\kappa$ — which is the logarithm of [Example 22.8](#)'s equation. At $L = 10$, $c = 3$ both give $\kappa = 0.2775645857$ · *checked*. The factor 2 in the jump is the $\hbar = 2m = 1$ units: $\partial_1^2 + \partial_2^2 = \frac{1}{2}\partial_X^2 + 2\partial_x^2$.

Worth doing by hand. Set $U \rightarrow \infty$ in the displayed equation *along the scattering branch*, where k_1, k_2 stay real and E stays bounded: the right-hand side tends to -1 , so $k_j L \in \pi + 2\pi\mathbb{Z}$ and the two electrons quantise as free spinless fermions on a ring with antiperiodic boundary conditions. The spin has not disappeared — the state is still a singlet — but it has stopped costing energy. That is the lattice face of the impenetrable limit of [Example 22.8](#). The branches with $E \sim U$ — the repulsively bound pair, $k_{1,2} = \pi \pm iq$ with $q \sim \ln(U/2)$, and the η -pair — are not hard-core states and the limit does not describe them.

Summary.

- *With labels the two-body problem has two answers* (22.3): $S(k)$ on the singlet, -1 on the triplet. Following the momenta, a crossing is the matrix $X(u) = R(-u)/(u+i) = -L(u)/(u+i)$ (22.4), with $X(0) = -P$, $X(u)X(-u) = 1$ and the Yang–Baxter equation.
- *Periodicity is a matrix condition* ([Theorem 22.2](#)): $(-1)^{N-1} e^{ik_a L} X_a B = B$, with X_a the ordered product of the $N-1$ crossings particle a makes on its way round. The $(-1)^{N-1}$ is the fermionic exchange sign; dropping it replaces the two-body equation by $e^{ikL} = -(u+i)/(u-i)$ and doubles the momentum at the sizes checked. The reduction of the $N!$ conditions to these N , on *one* vector needs the Yang–Baxter equation twice, and its factor order is not free.

- The N conditions are one commuting family (Proposition 22.4, Proposition 22.5): $\mathcal{X}_a = -t(\xi_a)$ for the inhomogeneous Heisenberg chain (22.13), whose N sites are the particles and whose inhomogeneities are their own momenta $\xi_j = k_j/c$. Lecture 18 solved it.
- The nested Bethe equations (Theorem 22.7): (22.16) for the N momenta and (22.17) for the M spin rapidities, with $E = \sum k_j^2$ — the spin rapidities carry none of the energy and act only as phase shifts. The quantum numbers have parities $I_j \in \mathbb{Z} + \frac{M}{2}$, $J_\alpha \in \mathbb{Z} + \frac{N-M+1}{2}$, and a wrong parity is invisible in the logarithmic form. At $N = 2$ they reduce to Lieb–Liniger in the singlet and to free fermions in the triplet (Example 22.8).
- Hubbard (22.21): the Lieb–Wu equations (22.22)–(22.23) are the same system under $k \mapsto \sin k$, $c \mapsto U/2t$ (22.24) — derived at $N = 2$ in the tutorial and checked against exact diagonalisation at $L = 6$, where 20 of the 21 singlet states are regular and the twenty-first is the η -pair; the general derivation is cited, and the regular solutions are *not* complete — at isolated couplings a colliding pair leaves a state that is neither regular nor a descendant (Theorem 22.9). On an *even* ring the model has $SO(4)$ — with the η 's spectrum-generating for (22.21) itself and symmetries of $H - \frac{U}{2}\hat{N}$ (Remark 22.10) — and an R-matrix that is *not* of difference form, which is why it sits outside Lecture 17's machinery rather than inside it.
- The hierarchy: for κ species the same $R = u - iP$ works, it is sl_κ -invariant, and the $\kappa - 1$ levels of nesting are the nodes of $A_{\kappa-1}$, coupled by the Cartan matrix (22.25) — with the self-term $(b, \beta) = (a, \alpha)$ excluded, since $C_{aa} = 2$ would contribute a spurious -1 , and with the source only at the node the particles attach to. For $\kappa \geq 3$ the auxiliary monodromy has $\kappa - 1$ creation operators, which is where the word *nested* comes from.

Tutorial. Exercise Sheet 22.

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